Computational Fluid Dynamics Simulation and Analysis of Double Slope Solar Still

A Project Report

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DECLARATION

"We hereby declare that this project work is our own work and that, to the best of my/our knowledge and belief, it contains no other material or previously published material written by another person that has been accepted for the award of any other degree of the university or other institute of higher learning, except where due acknowledgment has been made in the project report".

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CERTIFICATE

*This is to certify that the project entitled "***Computational Fluid Dynamics Simulation and Analysis of Double Slope Solar Still***" is the original work carried out by* ***Aniket Chorghe (F008), Ameya Dhavale (F013), Vedant Mehta (F032) and Prahnesh Rao (F090)*** *the partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Mechanical Engineering, Mukesh Patel School of Technology Management, NMIMS University, Mumbai, during the semesters VII and VIII of the academic year 2022-23. The project work has been assessed and found to be satisfactory.*

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ABSTRACT

This report provides an overview of various types of solar stills and in water desalination and purification. The article discusses the advantages and limitations of passive single, double slope, and active solar still. It also examines the impact of solar radiation, water depth, and ambient temperature on solar still performance. The study highlights technological advancements in solar still design, including the use of phase transition materials and nanofluids, which could potentially improve their efficiency. The article concludes by emphasizing the need for further research to enhance solar still technology and make it more accessible in water-scarce regions. The CFD analysis examines Double Slope Solar Still performance in ANSYS Fluent software. The thermal efficiency and freshwater production of the solar still were evaluated, with the results indicating both thermal efficiency and freshwater production. The study also analysed the solar still temperature distribution and heat transmission processes. These findings may have implications for solar still design and optimization, especially in areas with strong solar radiation and limited freshwater.

List of Symbols

Symbol	Abbreviation
G	Radiation
τG	Transmissivity of radiation through glass
qk	Heat transfer due to conductions
qe	Heat transfer due to evaporation
qr	Heat transfer due to radiation
qga,r	Heat transferred due to radiation on glass surface
qc	Convection
qga,c	Convection due to glass surface
τgI(t)	Transmitted radiation
R'wI(t)	Partially reflected radiation

List of Abbreviations

	Abbreviation
CFD	Computational Fluid Dynamics
PCM	Phase Change Material
SWF	Steel Wool Fibre
PF	Pin Fins
SS	Solar Still
SWF	Steel Wool Fiber
FRP	Fibre Reinforced Plastic
EC	External Condenser
TPTC	Tracking Parabolic Thermal Collector
CBA	Collector Basin Area
FPC	Flat Plate Collector
TDSSS	Tracking Double Slope Solar Still
VMED	Vertical Multiple Effect Diffusion
ANN	Artificial Neural Network
HHO	Harris Hawks optimizer
PV/T	PhotoVoltaic Thermal
SWH	Solar Water Heater
VSBSS	V-Corrugated Single Basin Solar Still

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Chapter 1

INTRODUCTION

1.1 Solar Still

A solar still uses the Sun's heat to evaporate water, purifying it and distilling it to remove dissolved chemicals. They are used in areas where drinking water is scarce so that plants can absorb sunlight and produce clean water from polluted water. Stills are classified into two types: condensation traps and large-scale focused solar stills. In a solar still, impure water is held outside the collection and evaporated by sunlight passing through the clear collector. Clean water vapour condenses and drips into a tank on the cool inside surface.

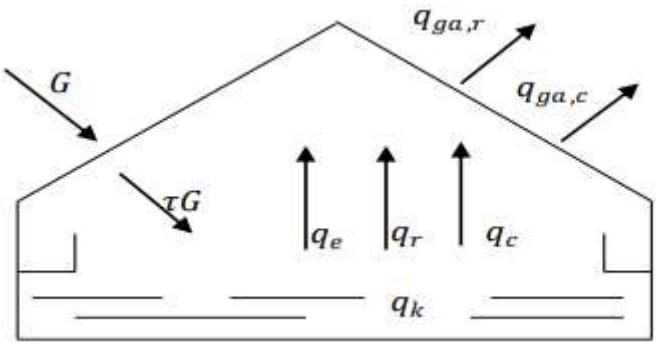

Figure 1. Double Slope Solar Still

Desalination is required in a solar still system to clean saline water, or seawater. The sun desalination method employs the same evaporation and condensation processes as the hydrologic cycle. Solar still uses the "green energy" energy transfer technique to convert dirty water into fresh water. In this technique, contaminated water is first added to the solar still basin. When sunlight penetrates a condensing cover, it heats the water to the point where it begins to evaporate. Water vapor then condensed on the slanted glass cover, accumulating contaminants and pure water droplets that finally dropped to the ground and were collected for use. As a result, the production of solar stills is solely dependent on incident sun radiation.

Solar distillation is separated into passive and active solar stills, with passive solar stills being more accessible and cost-effective. Active solar stills use external energy sources to boost evaporation and production.

1.2 Thermal Heat Storage Material

Thermal heat storage materials also commonly referred as Phase Change Materials (PCM) are compounds that can collect and release huge amounts of thermal energy during phase transitions such as melting or solidifying. These materials can be employed in a variety of applications.

A solar still is a device that uses sunshine to purify water by evaporating it and then condensing the vapor into liquid water. However, the effectiveness of solar stills can be limited by fluctuations in sunlight intensity and ambient temperature, which can impact the rate of evaporation and condensation. Latent heat storage materials can assist address this issue by storing extra thermal energy during periods of high sunlight intensity and releasing it during periods of low intensity or at night. This helps to maintain a more stable temperature within the solar still, boosting its efficiency and performance.

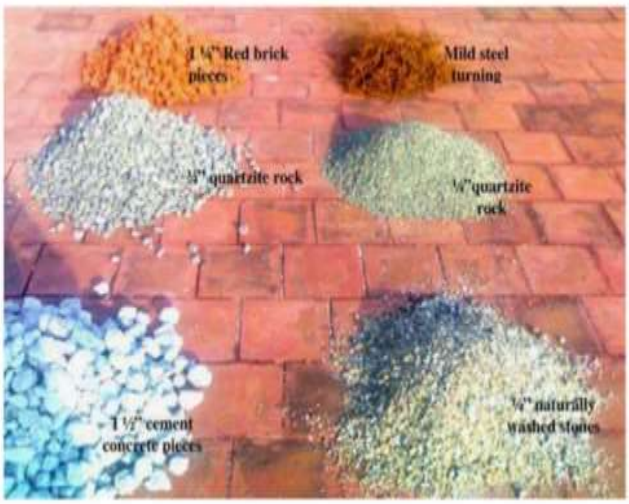

Figure 2. Thermal Heat Storage Material

Furthermore, the inclusion of latent heat storage materials can boost the overall water production capacity of a solar still by allowing it to operate for extended periods of time even in the absence of sunlight. This is especially important in locations with limited access to clean water, where solar stills can be utilized as a low-cost and sustainable method of water purification.

1.3 Computational Fluid Dynamics

Computational fluid dynamics is a popular numerical tool for analysing fluid flows and heat transfer processes. It enables the prediction and visualization of complex fluid behaviour and interactions with numerous structures. CFD can be used to solve a wide range of engineering problems, including aircraft design and environmental modelling.

CFD was used in this study to analyse the performance of a solar still with and without the introduction of phase change material (PCM). The goal was to see how PCM affected the thermal efficiency and fresh water production rate of the solar still. CFD simulations were performed with ANSYS Fluent, a robust and adaptable CFD software tool that is extensively used in academia and industry.

This project was chosen because CFD enables for a complete and quantitative examination of the fluid and heat flow phenomena inside the solar still, which would be difficult to obtain experimentally. CFD simulations allow for the investigation of various design and operational factors, as well as insights into the complex thermal behaviour of the solar still.

CFD simulations allow for the investigation of various design and operational factors, as well as insights into the complex thermal behaviour of the solar still. As compared to experimental testing, CFD is a cost-effective and time-efficient approach, making it an appealing tool for analysing complex fluid and heat transfer systems. Overall, CFD is an excellent method for better understanding the performance of solar stills, which can aid in their design and optimization for a variety of applications such as water desalination and purification.

Chapter 2

Literature Review

This research investigated the performance of a standard still, a modified still with leg fins absorber, and an external condenser. The study found that water absorption by leg fins and external condenser increased by 41.95%, 23.39%, and 11%, respectively. Leg fins with no air and external condensers increased water output by 32.18%. Additionally, T. Rajaseenivasan et al. (2016) discovered that the still's performance was affected by the basin's water depth, temperature, water-glass temperature difference, and exposure. Solar stills were used to evaluate hollow circular and square fins, and fin-incorporated square and round basins outperformed.

To improve PCM heat transmission, Saleh Abo-Elfadl et al. (2017) attached v-corrugated absorber plates to the storage tank and solar panels. Using two novel condensers, the researchers investigated the basin's media influence. The four-basin medium was filled with salt-saturated sand or steel fibres. The study found that solar energy was influenced by the saltwater medium and condensers. Condensers made of black steel-fibre glass increased production by 31%, and condensers by 35%. A 40-cm umbrella-cut aluminium plate condenser and saltwater generated 21% pure water.

M. H. Sellami et al. (2017) investigated traditional solar energy generation methods and found that black absorber plates on classic stills increased yield. Absorbent chemicals enabled nighttime distillation and increased the efficiency of solar stills. Bitumen, charcoal, and black ink improved solar panel efficiency by 25.35%, 18.42%, and 6.87%, respectively. Additionally, sunnier sponge rubber keeps you cool. The second law examines normal and modified solar stills, and the reticular porous layer boosts hourly output. The redesigned sponge is more energy-efficient than the solar system due to lower water levels.

Scientists used a basin and a five-millimetre-thick, seven-and-a-half-kilogram water layer to simulate sun radiation in a passive double-slope solar still. The water level in the basin necessitated the use of light black cotton fabric, light jute cloth sponge sheets, coir mate, and cotton scraps. Rectangular aluminium fins coated in cotton and jute cloth produced more and performed better than light black cotton fabric. According to the authors, theoretical and experimental parameters correspond.

K. Kalidasa Murugavel et al. (2010) created a single-basin double-slope solar still for testing energy storage. M. R. Rajamanickam et al. (2012) found that the north-to-south DS slope solar produced more distillate, evaporative heat transfer, and

condensate than the SS. Suresh Kumar Patel et al. (2020) designed an ultra-modified double slope solar still that captures and directs solar energy through transparent side walls to efficiently manufacture drinking water.

In 2018, Velmurugan et al. (2019) tested a double-slope, single-basin solar still. An outside glass reflector raised the temperatures of the ambient, basin, water, and vapor to 47.5°C, 80.4°C, 69.6°C, and 77.4°C, respectively. Heat storage material kept the water warm, while external reflectors quickly heated the modified conventional basin. In double-slope solar stills, heat storage, reflectors, and internal and external changes boosted distillate per hour. Conventional exceeded 1880 ml/m2/day by 58.57%. Analytical optimization of the inclination degrees of the glass cover's two facing sides was performed. This optimization projected annual solar energy transmission for latitude angles 24°-31.2° in the study by Velmurugan et al. (2019) on a double-slope, single-basin solar still.

Maheshwari et al. investigated a single basin double slope solar still using ANSYS CFX 14.0 to determine overall production. The overall output is highest in March and November. Gokilavani et al. conducted experiments and modeling on a single slope solar still. The geometry of the model was created in ANSYS Workbench 14.5 and then simulated in ANSYS CFX. The

The experimental work was compared to the simulated study, and the water temperature distribution over the glass surface was verified for November 27th and 28th, 2013. The result shows that the maximum temperature achieved inside the glass on 27/Nov/2013 is higher than on 28/Nov/2013.

Mahendren et al conducted a simulation study for double slope solar stills in order to improve their efficiency and productivity. The mathematical model and coding were completed using the MATLAB program to calculate the various heat fluxes in the still, determine the hourly output of the still, and determine the still's efficiency. Complete numerical analysis was performed, and various characteristic graphs were plotted in MATLAB. The Simulink toolbox was used to demonstrate simulation. Mfiles, which are ASCII text files, were used to calculate convective, evaporative, and radiative heat transfer rates. Nayak et al [9] performed thermal modeling and experiments on a double slope solar still.

A transparent acrylic sheet was used to design the modified double slope solar still, and an opaque fiber-reinforced plastic was used to impose the north wall of the solar still. The geometry of the model was created using the CATIA V5R18 software. The simulation and experimental results were nearly identical.

The experiment was carried out in Allahabad (25.23°N, 79.09° E) to evaluate the productivity of a solar still. The modified double slope solar still produced 16 kg of purified water from 25 kg of brackish water, forming it twice as much as the conventional solar still.

Tripathi and Tiwari used the solar fraction as a parameter to create a simulation model for thermal analysis of a single-slope solar still. The model's geometry was created using AUTOCAD 2000. Weather conditions in New Delhi (27.0238° N, 74.2179° E) were simulated. The solar fraction was estimated using a MATLAB program that calculated convective and evaporative heat transfer coefficients. In modeling, the following formula for the solar fraction (Fn) of a specific wall was still used.

Chapter 3

Working of Double Slope Solar Still

he solar radiation, after reflection and absorption by the glass cover is transmitted inside enclosure of the Solar Still. This transmitted radiation τgI(t) is further partially reflected R'wI(t) and absorbed α'wI(t) by the water mass. The attenuation of solar flux in water mass depends on its absorptivity and depth. The solar radiation finally reaches the blackened surface where it is mostly absorbed. After absorption of solar radiation at the blackened surface, generally known as the basin liner, most of the thermal energy is convected to water mass and a small quantity is lost to the atmosphere, by conduction. Consequently, the water gets heated leading to an increased difference of water and glass cover temperatures. There are basically three modes of heat transfer, radiation qrw, convection qcw and evaporation qew from the water surface to the glass cover. The evaporated water gets condensed on inner surface of glass cover after releasing latent heat. The condensed water trickles into channels provided at lower ends, under gravity. Collected water in the channel is taken out for further use. Thermal energy received by glass cover through radiation, convection and latent heat, is lost to the ambient by radiation and convection.

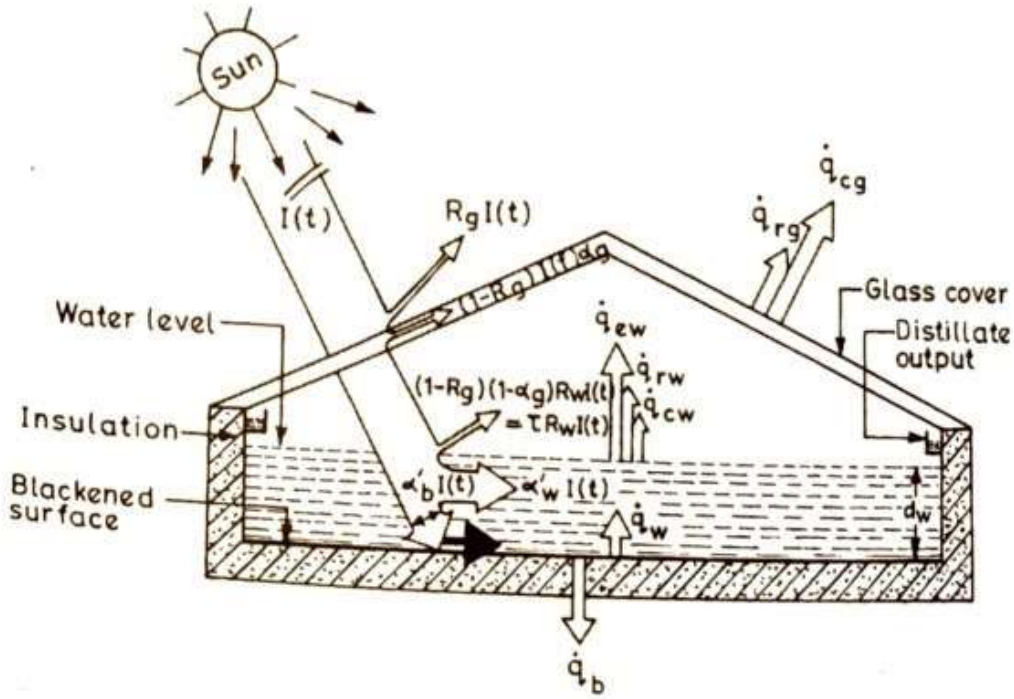

Figure 3. Energy flow diagram of double slope solar still

Chapter 4

Detail Design of Double Slope Solar Still

For adiabatic walls of thickness 0.05 m we have used wood as an insulating material. An adiabatic wall does not let heat in or out of a thermodynamic system and for maximum and efficient absorption we have used copper plates.

We are using wood as a material for adiabatic walls because it is cost effective and readily available almost everywhere. The thermal conductivity of wood is generally within the range of 0.04–0.12 W/m-K.

We have used glass for cover (thickness 0.004) because it has good transmittance, and wetting properties for minimising the loss of solar radiations during the condensation process. For adiabatic walls (thickness 0.05 m) we have used wood as an insulating material. An adiabatic wall does not let heat in or out of a thermodynamic system and for maximum and efficient absorption we have used copper plates.

An isometric view of Double slope solar still:

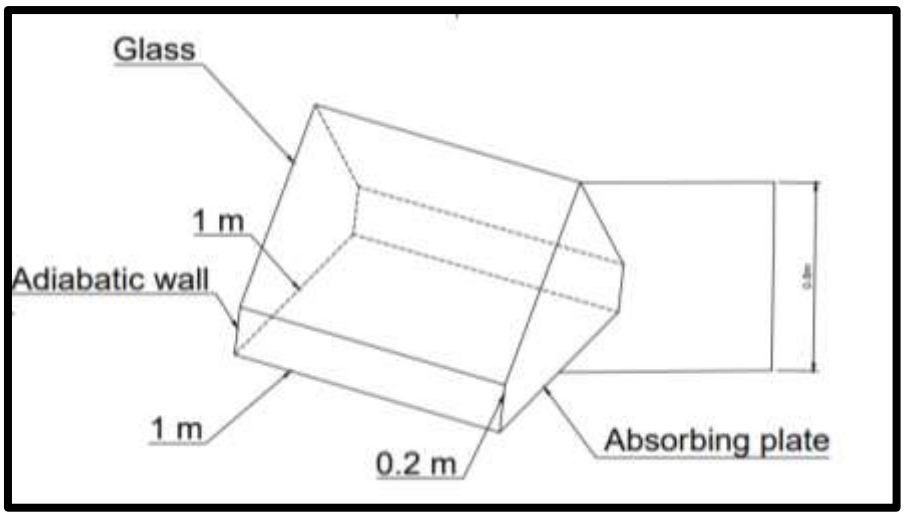

Figure 4. CAD Model of Double Slope Solar Still

4.1 Material Selection

The materials utilized in solar still systems will change because of the many various factors involved, such as the availability of these resources in the local area, as well as other financial and human considerations. The major goal is to emphasize the minimum requirements that must be met in order to ensure a long, usable, and productive life with little maintenance when operating in field conditions.

Clearly, as new construction materials are developed, consideration must be given to the possibility that these materials will be used in solar distillation applications.

When it comes to determining when these more sophisticated materials will be ready for widespread use of this technology, economic considerations will be critical. In the end, the most essential goal in the optimal material selection for a solar still is to provide clean water for people to utilize primarily as drinking water and cooking water. This is the main goal in the ideal material selection. As a result, it must be of the highest possible quality to prevent disease spread.

4.2 General Specifications of Solar Stills

There are some fundamental prerequisites that absolutely have to be satisfied. In general, the unit needs to be able to satisfy the conditions listed below:

1. It needs to be simple to put together in the field.
2. It needs to be able to be packaged up to cut down on the expense of shipping.
3. It should be lightweight for ease of handling and shipping;
4. It must have an effective life, with normal maintenance, of 10 to 20 years;
5. The still must be equipped with access ports for ease of maintenance;
6. It should not require or depend upon external power sources;
7. It should also have access ports for easy maintenance;
8. It should also have an adequate capacity.
9. It must be manufactured in such a way that the maximum solar still component size can be directly related to economic shipping dimensions as specified by goods carriers.

4.2.1 Transparent Cover

This serves to cover the equipment and transmit solar radiation to the interior. Properties required are listed below:

1. The material should be able to endure the effects of various environmental factors such as wind, sunlight, rain, and dust.
2. The material should allow over 85% of shortwave solar radiation (0.3 to 3.0 microns) to pass through, preferably more.

3. The material should block long-wave radiation (over 3.0 microns) by being nearly opaque.
4. The material should not absorb a lot of water, as it may be used for catching rainfall and as a condenser.
5. The material's solar reflectivity at normal incidence should not exceed 10%.
6. The material should have low solar absorptivity, especially if it is also to be used as a condenser.
7. The material should have high heat capacity to keep the cover temperature low.
8. The material's properties should remain constant over time and exposure to the elements.
9. The material should not attract dust particles on the outside surface by having electrostatic properties.
10. If the cover is to be used as a condenser, the material should be able to withstand high temperatures of up to 80°C and high relative humidities on one side while also enduring low humidities on the other side.
11. The cover should be able to resist small animal damage.
12. The cover should be able to withstand high wind loads of up to 45 m/sec.

4.3 Material Properties

Table 1. Material Properties of Walls in Absorber plate

Material	**Density (kg/m3)**	**Specific heat capacity J(Kg-k)**	**Thermal conductivity W/(m-K)**
Low iron tempered glass	2530	840	1.2
Copper	8960	385	386
Wood	720	1700	0.12

4.4 Designing of Solar Still

For adiabatic walls (thickness 0.05 m) we have used wood as an insulating material. An adiabatic wall does not let heat in or out of a thermodynamic system and for maximum and efficient absorption we have used copper plates.

We are using wood as a material for adiabatic walls because it is cost effective and readily available almost everywhere. The thermal conductivity of wood is generally within the range of 0.1–0.2 W/m-K.

We have used glass for cover (thickness 0.004) because it has good transmittance, and wetting properties for minimising the loss of solar radiations during the condensation process.

Figure 5. Isometric view of double Slope solar still

Chapter 5

Analysis of Double Slope Solar Still

5.1Governing Equations

5.1.1 Energy Equation

$$\rho Cp(\partial T/\partial t + u.\nabla T) = \nabla.(k\nabla T) + Q$$

where T is the fluid temperature, ρ is the fluid density, Cp is the specific heat capacity, k is the thermal conductivity, u is the fluid velocity, t is time, ∇ represents the gradient operator, and Q is the volumetric heat source.

The first term on the left-hand side represents the rate of change of thermal energy with time and the convective transport of thermal energy due to fluid motion. The second term on the right-hand side represents the conductive transport of thermal energy due to temperature gradients. The third term represents any volumetric heat sources or sinks that may be present, such as solar radiation or internal heat generation.

5.1.2 Continuity Equation

$$\partial \rho/\partial t + \nabla. (\rho u) = 0$$

where ρ is the fluid density, t is time, u is the fluid velocity, and ∇ represents the gradient operator.

The first term on the left-hand side represents the rate of change of mass with time, while the second term represents the convective transport of mass due to fluid motion. In your CFD analysis of the double slope solar still, the continuity equation will be solved to ensure that mass is conserved throughout the simulation.

5.1.3 Navier - Stoke equation

The Navier-Stokes equation describes the motion of fluids and can be expressed in differential form as:

$$\rho(\partial u/\partial t + u.\nabla u) = -\nabla p + \mu\nabla^2 u + \rho g$$

where ρ is the fluid density, u is the fluid velocity, t is time, p is the fluid pressure, μ is the fluid viscosity, g is the gravitational acceleration, and ∇ represents the gradient operator.

5.2 Analysis and Setup

Computational Fluid Dynamics (CFD) is a valuable tool that has been used to analyse and investigate the fluid flow pattern of moisturized air and temperature variation near the walls. For visualizing the flow field, any location computational fluid dynamics can prepare the design, get the controlling step, and guide. In the study, computational fluid dynamics is utilized to create geometry, and the simulation result is obtained by using ANSYS 23 student version.

5.2.1 Grid Generation

Meshing is a crucial stage in numerical simulation that entails partitioning the computational domain into small discrete pieces or cells to simulate the physical behaviour of the system under consideration. Meshing is especially crucial in the case of a double slope solar still in order to precisely capture the heat transfer processes that occur between the absorber plate and the glass cover. To discretize the computational domain in this situation, a body sizing element size of 20mm was selected. Additionally, the absorber plate and glass were refined to guarantee that the mesh resolution was sufficient to capture the temperature and heat flux fluctuations that occur inside these zones. Number of nodes are 69450 and number of elements are 61985.

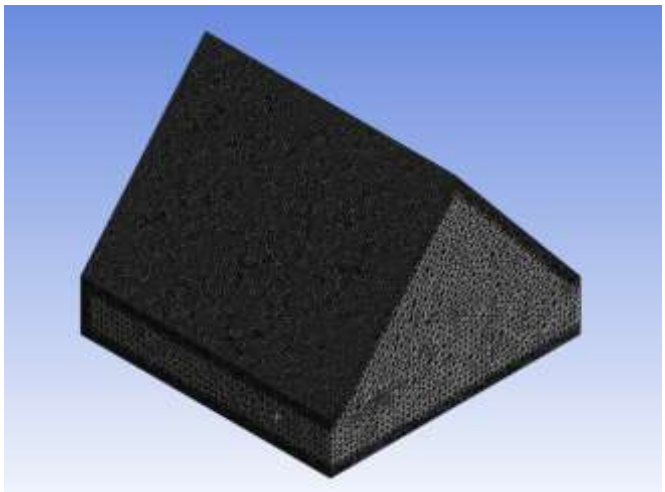

Figure 6. Unstructured Mesh of Double Slope Solar Still

5.2.2 Boundary Conditions

The boundary conditions used in the CFD simulations of the solar still with and without phase change material were essential for accurately modelling the physical behaviour of the system. The boundary conditions specified the velocity and temperature of the air and water entering and exiting the solar still, as well as the wall boundary conditions and the radiation models used to account for the effects of solar radiation on the system. These boundary conditions were carefully chosen to replicate the real-world conditions of a solar still operating in a typical environment. Properly setting up the boundary conditions in the CFD simulations is crucial for obtaining accurate and reliable results that can be used to optimize the design and operation of the solar still for maximum efficiency and effectiveness.

Table 2: Boundary Conditions for the simulation process of Double Slope Solar Still

Domain	**Domain Type**	**Location**	**Boundary Type**	**Boundary Details.**
Solid	Cell	Absorber Plate	Wall	Stationary wall with heat flux
		Glass Roof	Wall	Fixed wall Temperature
		Sidewalls	Wall	Adiabatic
		Front and Back Wall	Wall	Adiabatic

5.2.3 Solver Setup

The convergence criteria used in the computational fluid dynamics simulations of the solar still with and without phase change material were critical for ensuring that the solutions obtained were accurate and reliable. The convergence criteria specified the maximum number of iterations allowed, as well as the tolerance values for the residuals of the governing equations. The residual is a measure of the difference between the actual and predicted values of the variables being solved for, and a lower residual indicates a more accurate solution.

The numerical schemes used in the CFD simulations of the solar still with and without phase change material were critical for accurately solving the governing equations of the system. The numerical schemes specified the discretization methods used for the various terms in the governing equations, such as advection, diffusion, and source terms. The numerical schemes used in this study were chosen

to balance accuracy with computational efficiency. A second-order upwind scheme was used for the advection terms, while a second-order central differencing scheme was used for the diffusion terms and second order scheme was used for energy and momentum equation. A first-order implicit scheme was used for the volume fraction, as it provided the necessary stability for the simulations.

In the fluent, physics and boundary conditions are specified to solve the continuity equation, momentum equation, and energy equation at all the boundaries. The RNG k-ε viscous model was used, which predicts the spreading rate of flow and better water production performance. In the model, the near-wall condition was treated using 'Standard Wall Function". Adhesion forces are taken for producing droplets inside the solar stills. No-slip wall boundary conditions were specified for both gas phase and liquid phase. Computational fluid dynamics simulation had a run time of 5 hours double slope solar still cases.

The location of Experiment was Mukesh Patel School of Technology Management and Engineering. Operating pressure is 1.01 bar and initial temperature was 303K.

Table 3: Solver Setting for Double Slope Solar Still

Function	**Specification**	
General	Type	Pressure-based
	Space	3D
	Time	Transient
	Gravity	-9.8 m/s^2 (Y-axis)
Model	Multiphase	VOF Model, Implicit Scheme
	Viscous	RNG k- ε Turbulence Model
	Radiation	Solar Ray Tracing and Solar Loading Inputs: 19.1101° N, 72.8376° E Day: 21.02
Phases	Primary	Water Vapor
	Secondary	Water Liquid

The VOF (Volume of Fluid) multiphase model is a popular method for simulating the behaviour of two or more immiscible fluids in a given domain. In this particular case, the two phases being considered are water vapour and water liquid, with water vapour being the primary phase and water liquid being the secondary phase. The surface tension between the two phases is set to a value of 73 dyne/cm2, which reflects the degree of attraction between the two fluids. Additionally, the mass transfer between the water liquid and water vapour is modelled using the evaporation-condensation approach, specifically using the Lee model. This involves the transfer of mass between the two phases as water vapour is evaporated from the liquid phase and then condenses back into liquid form when it comes into contact with a cooler surface. By incorporating these various factors, the VOF multiphase model can provide a detailed and accurate representation of the behaviour of these two fluids and their interaction within the domain being considered

Chapter 6

Results and Discussion

6.1 Simulation Result

ANSYS FLUENT 23 student software was used to carry out the results using double precision and run on parallel computers. In Simulation results, the temperature distribution, velocity, water volume fraction, solar heat flux, and water production of both solar stills are examined and predicted the temperature and water production of both cases for 9am to 4pm for double slope solar stills.

6.1.1 Temperature Distribution

To visualize the temperature distribution, volume rendering and contour are used to create color-coded images for simulation of 12-1 pm where we can observe behaviour of double slope solar still. In the images, the blue colour represents the minimum temperature while the red colour represents the maximum temperature. The analysis revealed that for the double slope solar still, the maximum temperature observed at 11am was 343 K, while the minimum temperature was 350.55 K. These results provide valuable insights into the behaviour of the solar still and can be used to optimize its design and performance.

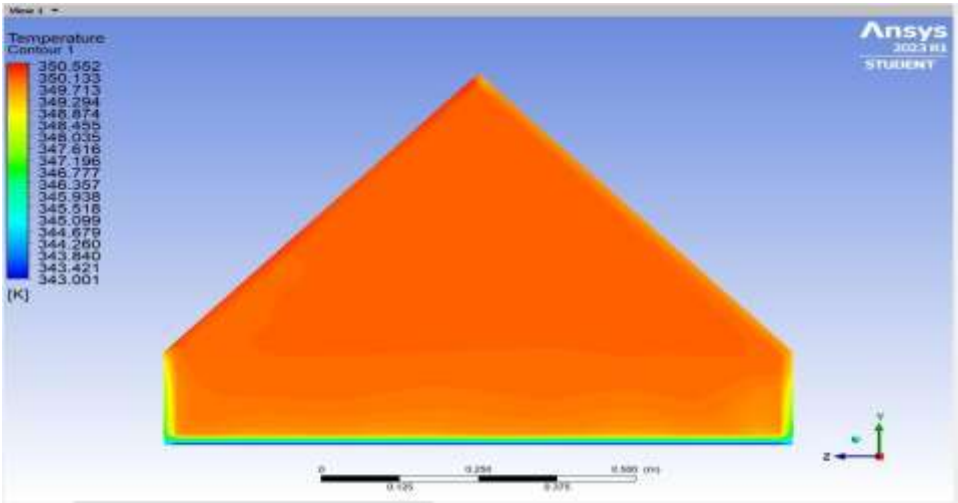

Figure 7. Temperature distribution inside solar still

Figure 6 provides a detailed view of the temperature distribution inside a double slope solar still from 12 noon to 1pm. The contour plot shows that the maximum temperature observed during this period was 345.72 K, which was located on the glass roof of the still. The minimum temperature observed was 343 K, located on the bottom of the solar still. This temperature distribution provides insights into the

heat transfer mechanisms at work within the still and can be used to optimize its design for maximum efficiency.

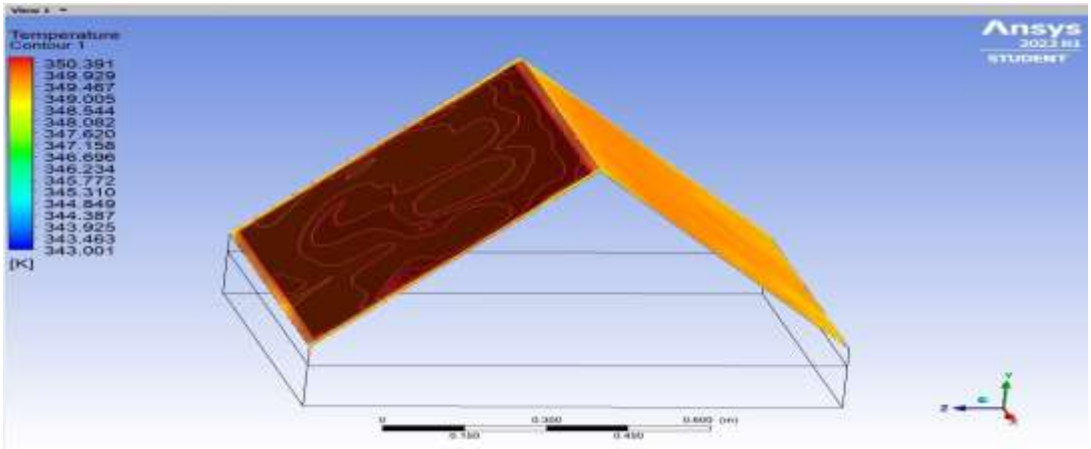

Figure 8. Temperature distribution on glass roof

Figures 7 and 8 complement this analysis by providing a more comprehensive view of the temperature distribution within the still. Figure 7 shows the temperature distribution on the glass wall of the still, while Figure 8 provides a volume rendering of the temperature distribution inside the still. These images provide valuable information about the behaviour of the fluid and heat transfer mechanisms within the still, and can be used to optimize the design of the solar still for maximum efficiency and output.

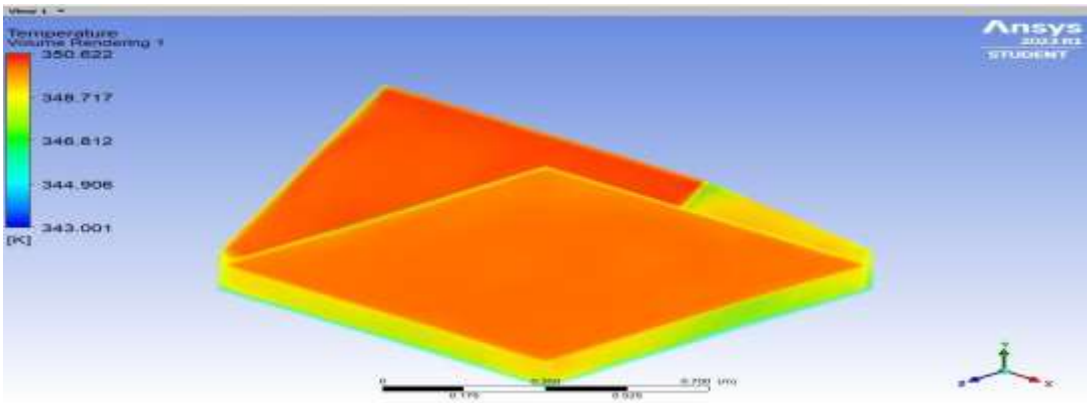

Figure 9. Volume Rendering temperature distribution inside still

Temperature of water liquid is observed to be 348 K average. Figure 9 is volume temperature of water liquid inside solar still.

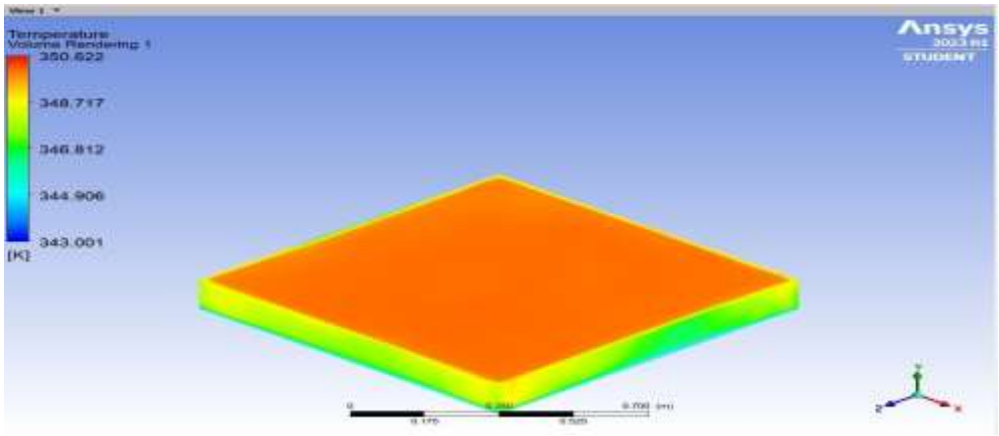

Figure 10. Volume Rendering of Water Temperature

Table 4. Results for hourly variation of water, glass temperature and water yield

Time (hr)	**Water Temperature (°C)**	**Glass Temperature (°C)**	**Water Yield (kg/m^2-hr)**
09:00	45.95	51.65	0.1527
10:00	50.59	56.92	0.1397
11:00	53.73	63.47	0.1487
12:00	54.94	67.05	0.2856
13:00	54.29	72.47	0.3981
14:00	52.05	69.62	0.5437
15:00	47.98	65.15	0.5023
16:00	42.45	63.69	0.4262

The temperature of both solar stills is expected to be between 08:00 and 17:00 hr. Figure 10 depicts the temperatures of both solar stills at various hours. The maximum temperature was reached at 12:00 p.m. in the double slope solar still.

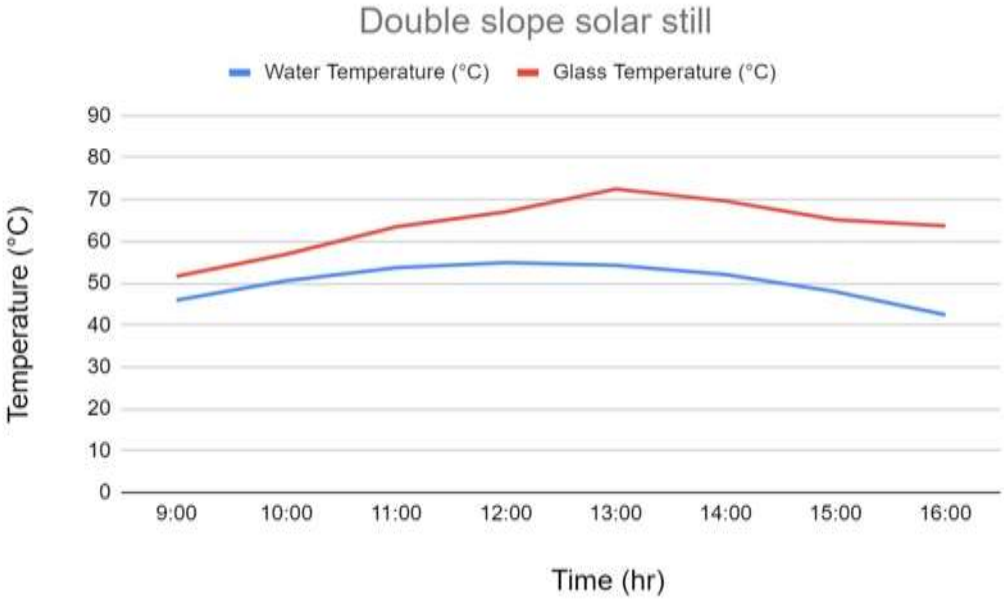

Figure 11. Hourly water and glass temperature of double slope solar still

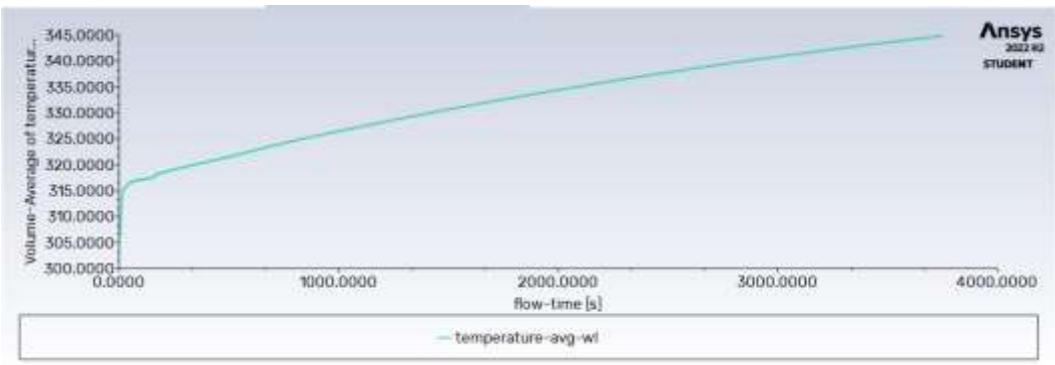

Figure 12. Temperature variation in graph for 12-1 pm

Figure 11 presents graphs depicting the hourly variation in water temperature and glass temperature within a solar still from 9am to 4pm and figure 12 represents graph for temperature variation for 12-1pm. These graphs provide valuable information about the performance of the solar still over time. The water temperature graph shows how the temperature of the water within the still changes throughout the day, with the highest temperature observed in the late afternoon. The glass temperature graph shows how the temperature of the glass changes over time, with the highest temperature observed in the early afternoon. By carefully analysing these graphs, it may be possible to optimize the design and operation of the solar still for maximum efficiency and output.

Figure 13 presents a graph depicting the mass of water evaporated from a solar still over time. This graph provides valuable information about the efficiency of the still in converting solar energy into water vapor. The graph shows that the mass evaporated is highest around 11 am, with a value of 0.1451 kg/m2-hr. This indicates that the solar still is most efficient at converting solar energy to heat and water vapor during this time period.

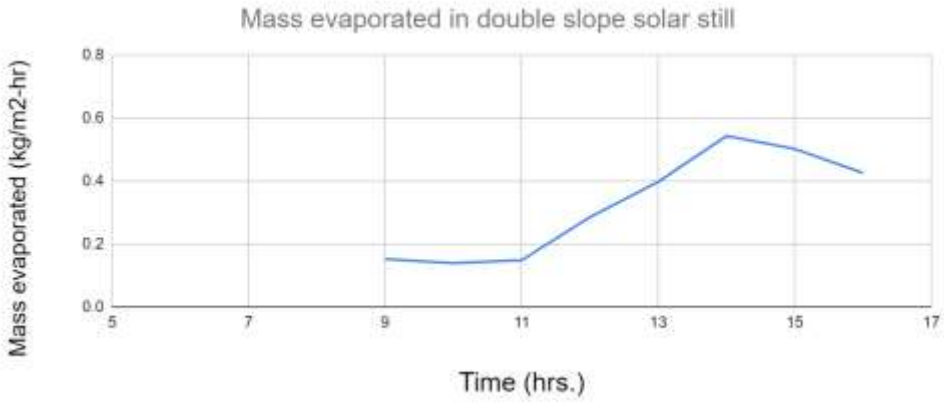

Figure 13. Hourly graph Mass water evaporated graph

6.1.2 Volume fraction

The volume fraction shows how much of a given volume a certain fluid, like water or air, takes up. This parameter is an essential aspect of the analysis as it provides information on the distribution of fluids and their behaviour within the system. By examining the volume fraction of the fluid, we can determine the regions where the fluid is stagnant or flowing and assess the performance of the solar still in terms of its ability to evaporate water and condense it into a separate compartment.

6.1.2.1 Water Liquid

Water liquid volume fraction represents the fraction of the total volume occupied by the liquid phase of water within the still. In the still, the liquid phase of water is heated by the incoming solar radiation, causing it to evaporate and form water vapor. The amount of liquid water presents in the still directly affects the overall heat transfer process and the efficiency of the still. Therefore, in the results section of the report, it is essential to present a detailed analysis of the water liquid volume fraction, highlighting its variations over time and space. This analysis can provide valuable insights into the behaviour of the still and its potential for optimization.

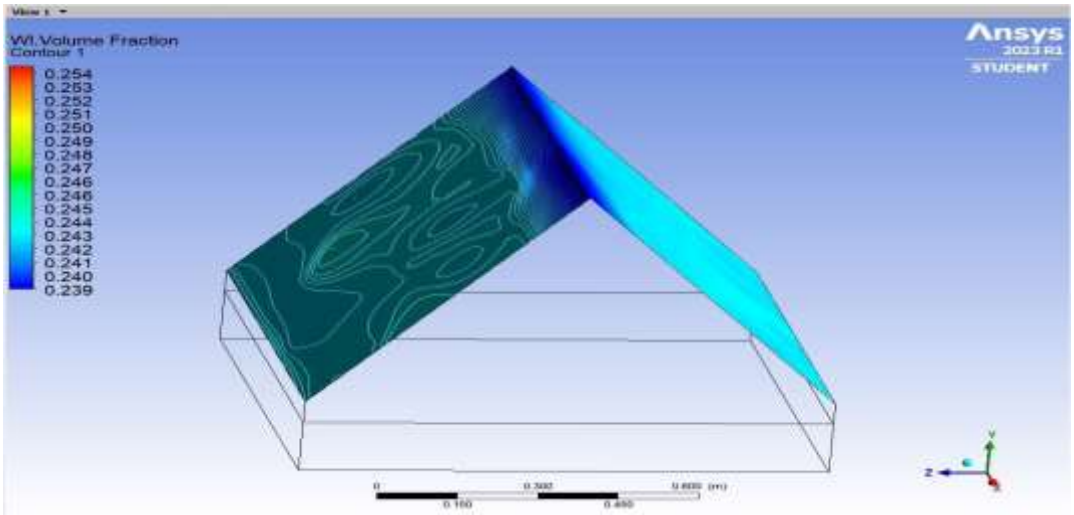

Figure 14. Volume Fraction of Water liquid Fraction on glass roof

The blue colour shows less amount of the volume fraction of water, while the red colour shows more water volume. The volume fraction of droplets is increasing downward on the glass. The maximum water volume fraction on the glass is 0.254 and a minimum of 0.239.

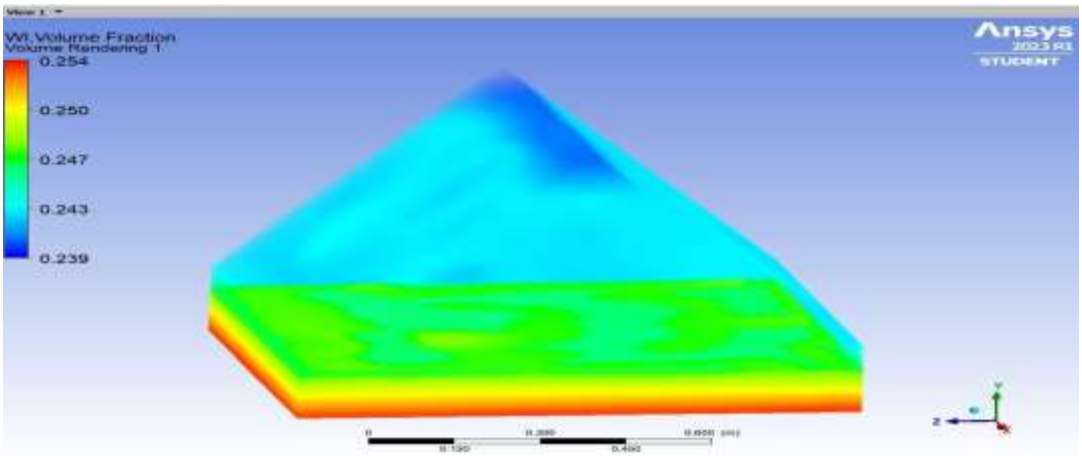

Figure 15. Volume Rendering of Water Volume fraction

6.1.2.2 Water Vapor

The water vapor volume fraction represents the fraction of the total volume occupied by the water vapor phase within the still. The water vapor phase is formed when the liquid phase of water evaporates due to the heating effect of solar radiation. The water vapor phase is responsible for the mass transfer process in the still, and its behaviour is critical to the overall efficiency of the still. This analysis can help identify areas where the mass transfer process could be improved, and it can also provide valuable insights into the behaviour of the system under different operating conditions.

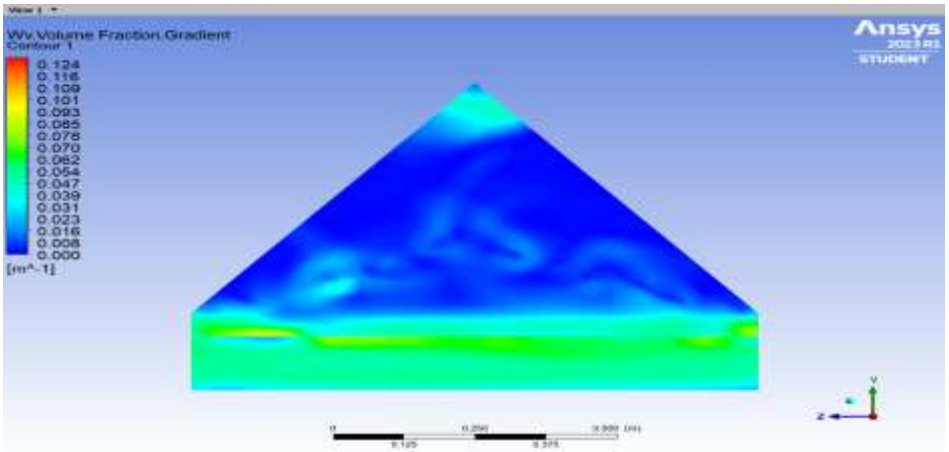

Figure 16. Water vapor volume fraction gradient inside solar still

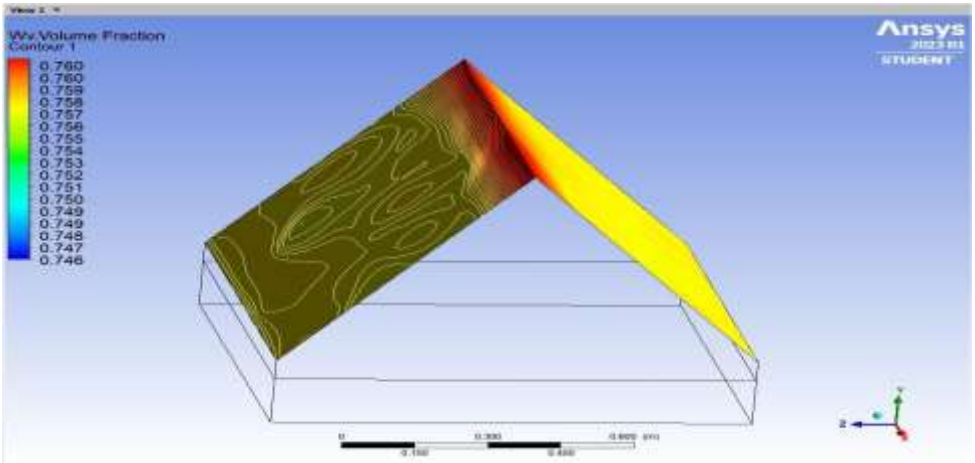

Figure 17. Vapor volume fraction on glass roof

Figure 16 and Figure 17 provide critical information about the distribution of water vapor within a solar still. Figure 15 presents a contour plot of the water vapor volume fraction on the glass roof of the still, while Figure 16 shows a volume rendering of the vapor fraction within the still. These images show that the volume fraction of water vapor varies significantly throughout the still. The range of vapor fraction observed was from 0.761 at the top of the still on the glass roof to a minimum of 0.746 at the bottom absorber plate. This distribution of water vapor provides insights into the heat and mass transfer mechanisms at work within the still and can be used to optimize its design for maximum efficiency.

```
                                wv
            Area-Weighted Average
                  Volume fraction
-------------------------------- --------------------
                   absorber_wall           0.73827799
       absorber_wall-1:external                    0
             adiabatic_wall-air           0.74148264
    adiabatic_wall-water-liquid           0.73899093
                          glass           0.74192695
      interior-air-water-liquid           0.74027196
                ---------------- --------------------
                             Net           0.61625737
```

Figure 18. Water vapour Weighted Average volume fraction

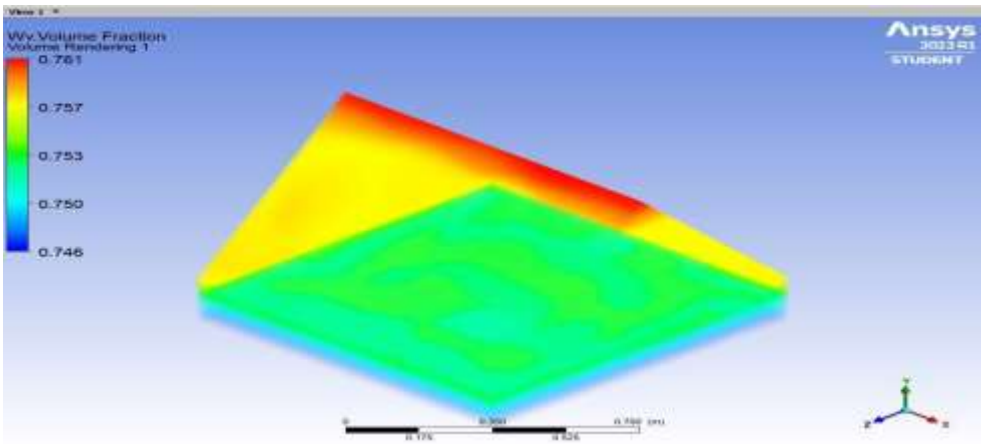

Figure 19. Volume rendering vapor volume fraction

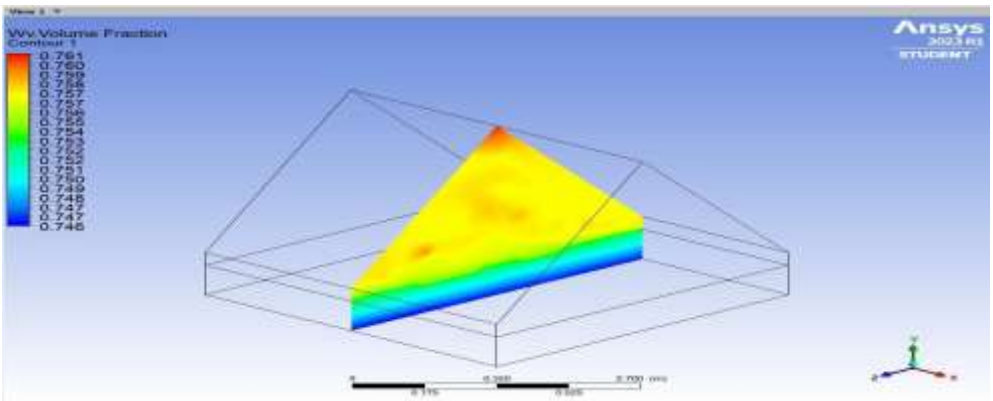

Figure 20. Volume rendering of water vapour volume fraction

6.1.3 Velocity

The vapour phase velocity vector of the double slope solar still is shown in Figure 18. First, the water changed into vapour and started to move upward due to the buoyancy effect. The vapour phase follows a circular path from the bottom to the glass surface. Vapour condenses on the glass and changes into water droplets, which move downward on the glass surface. This process takes place in both solar stills, which show the heat transfer mechanism. There is a difference in vapour velocity in both solar stills, which is more near the bottom in single slope solar still than in the double slope solar still. The maximum velocity of vapour is only 0.70289 m/sec, this difference of vapour velocity influences water production.

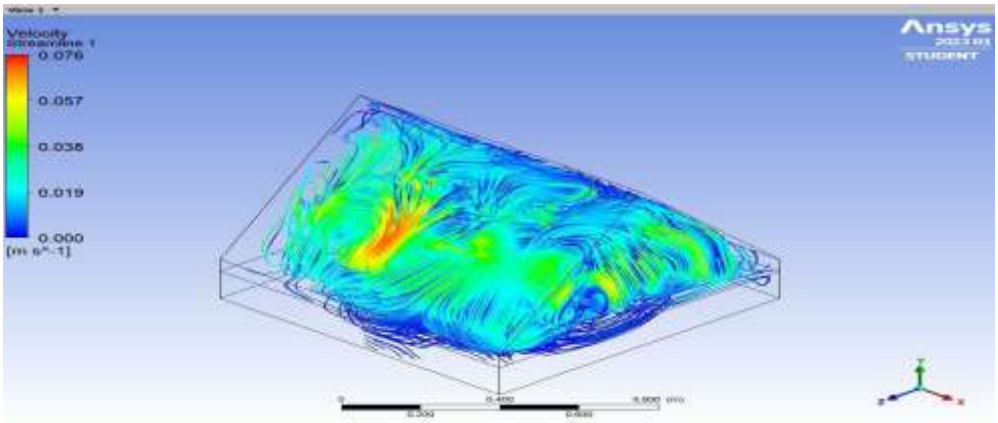

Figure 21. Streamline velocity inside solar still

6.1.4 Solar Intensity

The solar intensity represents the amount of solar radiation that the still receives, and it is a function of factors such as time of day, latitude, and cloud cover. In the result section of the report, it is important to include a detailed analysis of solar intensity and its impact on the performance of the double-slope solar still. The data on solar intensity can be used to figure out how much energy can be put into the still and also to figure out how well the still works in different weather conditions. By looking at the data on solar intensity, it is possible to see where the design and operation of the double-slope solar still could be improved.

Table 5. Solar Intensity in Double Slope Solar Still

Time (hrs.)	Solar Intensity (W/m^2)
9	525.9
10	680.78
11	735.55
12	810.13
13	790.09
14	710.89
15	598.51
16	430.83

When Solar energy incidents on the glass surface increase the inside temperature of the solar still and converts water into vapor. Figure 22 is graph that shows solar intensity received by double slope solar still for time 9am to 4pm. The solar intensity is observed is to be gradually increased from 525.0 W/m2 to going maximum at 12 pm to 790 W/m2 then decreasing again to 430.82 W/m2.

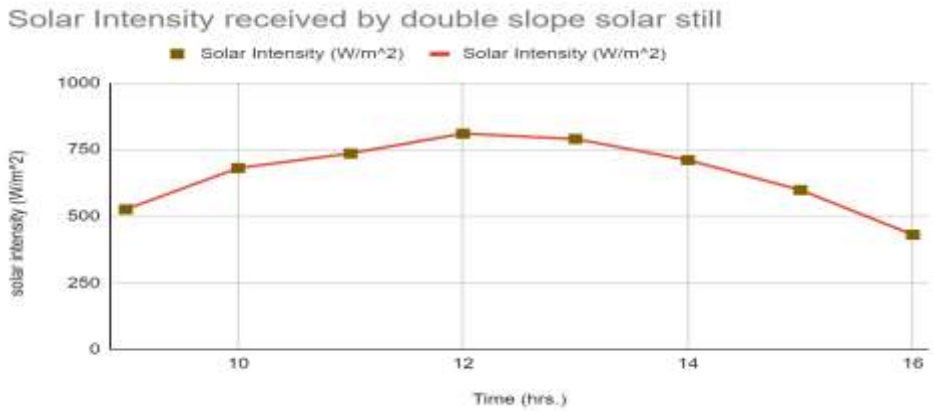

Figure 22. Variation of Solar Intensity with time of day

Chapter 7

7.1 Conclusion

In this project, we conducted a thorough literature review on solar stills and their various designs. We found that solar stills have been used extensively in arid and remote regions for their ability to produce potable water from saline or brackish water sources.

Following the literature review, we focused on the design and analysis of a double slope solar still using Computational Fluid Dynamics (CFD) simulations. The double slope solar still was chosen for its efficiency in producing distilled water, even under less favourable weather conditions.

Our analysis revealed that the temperature of the still had a significant impact on the water yield, with higher temperatures leading to higher water production. The water yield also varied throughout the day, with the highest yield observed during midday. The volume fraction of water vapor was highest near the top surface of the still, indicating that proper ventilation was necessary to remove the vapor and improve the efficiency of the still.

Overall, our findings suggest that the double slope solar still has the potential for efficient water production, even under less favorable weather conditions. However, further optimization is needed to improve its performance and maximize its water production efficiency. Our study provides a foundation for future research on the design and optimization of solar stills, with the ultimate goal of developing sustainable and cost-effective methods for producing potable water.

7.2 Future Scope

The CFD simulation of a double-slope solar still with latent heat storage material in this study has interesting implications for sustainable water and energy production. These avenues of research provide exciting possibilities for future developments in solar still technology. The following future scope points offer relevant research directions:

- Scale-up and field deployment: The current research project is centred on the CFD study of a small-scale solar still. However, the design must be scaled up and tested in the field to evaluate the performance of the solar still under real-world operating conditions. This would necessitate the design and construction of a larger solar still, as well as testing it in various climatic situations.
- PCM integration: The usage of PCMs with different melting temperatures can be researched in order to maximise the system's thermal storage capacity. This could help to increase the overall efficiency of the system and make it acceptable for extended operation durations.
- Enhancement of heat transport: The current study effort assumes uniform heat transfer throughout the system. Heat transfer improvement approaches such as Nano fluids or flow modifiers, on the other hand, can be investigated to improve heat transfer between the PCM and the water. This could improve the system's overall efficiency.
- Finally, you can compare the performance of the double-slope solar still to that of other solar still designs such as single slope, basin, or stepped designs. This would aid in determining the strengths and limitations of each design and determining the best design for a specific application.
- Nanomaterial integration: Nanomaterials like carbon nanotubes or graphene can be used into the design of a solar still to increase its thermal qualities. These materials have a high thermal conductivity and can be utilised to facilitate heat transfer between the solar still's various components.

Chapter 8

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Appendix A - **Assessment Criteria for Bachelor of Technology in Mechanical Engineering - Final Year Project**

Title of the Project: Computational Fluid Dynamics of Double Slope Solar Still			
No.	Criterion	Explanation of the Criterion	Check
1	Analysis	*Detailed summary of design techniques, theoretical derivations or empirical equations used to arrive at certain conclusions*	✓
2	Codes	*A system of design principles or rules*	
3	Constructability	*A degree to which knowledge and experience provide insight into achieving project goals under resource constraints prior to construction*	
4	Cost	*Cost may be one of acquisition, in which case the amount of money expended to acquire or to construct*	
5	Ergonomics	*Application of psychological &physiological principles to the design of products, processes, and systems.*	
6	Extensibility	*The ability of a design to take future additions in functionality into consideration*	✓
7	Functionality	*The quality of being suited to serve a set of purposes well*	✓
8	Maintainability	*The ease of maintenance and servicing*	✓
9	Manufacturability	*General engineering practice of designing products in such a way that they are easy to manufacture.*	
10	Policy	*A policy is a statement of intent, and is implemented as a procedure or protocol.*	
11	Regulations	*Abstract concept of management of complex systems according to a set of rules and trends.*	
12	Schedule	*Basic time-management tool, consists of a list of times at which possible tasks, events, or actions are intended to take place*	
13	Standard	*A required or agreed level of quality or attainment.*	
14	Sustainability	*Avoidance of the depletion of natural resources in order to maintain an ecological balance.*	✓
If the project satisfies any of the above criteria, please check (√) the category.			